

Smart Labs with Zonal Control using Machine Learning

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Abstract— Modern laboratory in universities nowadays rely heavily on digital infrastructure which consume high energy and causes a lot of waste in energy consumption due to irregular occupancy. Conventional automation system techniques are made to avoid this energy waste crisis such as using basic motion sensors, using systems on a fixed schedule but this leads to often failing to adapt real usage patterns within laboratory spaces. This paper proposes a Smart lab automation system that uses IoT (Internet of Things) hardware with machine learning with computer vision to enable zone wise intelligent lighting control. The proposed system uses a Raspberry pi 5 as an edge computing unit to control the automation system by processing real time video streams, where human presence is detected using a YOLO based model. When a human presence is detected the lights in the specific occupied zone are dynamically controlled by the GPIO pins on the Raspberry pi 5 by relay modules. This ensures that only required areas are illuminated. A web based monitoring and control system is designed using Flask and Socket IO which allows users to observe system status and manually control automated system decision whenever necessary. The experimental evaluation of the proposed system shows an energy reduction ranging from 35% to 70% with good response time and operational accuracy compared to traditional systems and conventional sensor based automation systems.

Keywords— Smart Lab Automation, IoT, Raspberry Pi, YOLO, Computer Vision, Energy Optimization, Zonal Control.

I. INTRODUCTION

These days everything on the university is digitalized and due to this there is a high energy consumption in the labs due to lights remain on, computers turned on when there is no occupant present in the lab. The traditional automation systems such as using systems on fixed schedules or sticking motion sensors which does work theoretically but is incapable to keep up with real life patterns of people movement and its utilization of the space.

Several studies report that conventional laboratory setups waste significant electrical energy due to the absence of adaptive control mechanisms [1],[10]. Traditional

automation techniques such as fixed schedules which operate on predefined timing and do not respond to real-time occupancy present [4]. Similarly, PIR-based motion sensors provide limited awareness and often fail to detect the stationary occupants in shared laboratory environments where occupancy varies dynamically across space and time [4],[9].

Recent advancements in the Internet of things (IoT) and embedded systems have enabled centralized monitoring and control of laboratory infrastructure. IoT based smart laboratory systems demonstrate enhancements in operational efficiency and management, although most implementations rely on rule-based logic and do not have any individual intelligent thinking capabilities [5],[11]. Therefore, these approaches fail to distinguish between operational and idle zones within a single laboratory space. The integration with machine learning with IoT introduces adaptive and data driven automation. Earlier research shows that ML enabled IoT systems can analyse usage patterns which optimizes resource allocation, and enable prognostic control [6],[7]. However, the application of such an intelligence to fine grained, zonal level lighting control in academic laboratories still remains limited.

The introduction to low-cost edge computing devices such as Raspberry pi 5 makes it possible to run deep learning lightweight models locally in real time, this eliminates the latency in response time and since it runs locally there are not much of privacy concerns associated with cloud-based processing. At the same time, computer vision models such as YOLO which helps in accurate human presence using camera feeds in real time offers a superior precision compared to traditional motion sensors [3]. This work proposes a smart laboratory automation system that combines YOLO based vision intellect, machine learning assisted zonal control, and Raspberry pi 5 based edge computing to achieve energy efficient, responsive, and scalable laboratory lighting automation.

II. LITERATURE REVIEW

Computer labs are a significant concern to energy consumption. Having automated systems the power use can be controlled by switching off idle computers and by allocating resources in optimum manner. The proposed

product is an IoT-based smart lab system that can be used to save 40% of energy used by operating smart strategies when used to control power[1]. More recent studies build upon the field of basic automation to include machine learning (ML) methods so that machines can make intelligent decisions. The IoT systems with ML can identify patterns of use, as well as forecast failures, and efficiently allocate resources [6], [7]. These strategies emphasize the possibility to evolve automation according to which the laboratory zones can be dynamically adjusted to real-time conditions in comparison with the predetermined timeline. Zonal control is an implicit idea that comes up in various researches, especially those that correspond to optimization of energy and management of smart infrastructure. IoT based laboratory systems are becoming more and more inclined toward segmented AI Writing Submission control over lab spaces, where various areas can be independently regulated in matters of lighting, power, and environmental control [5],[9]. Nevertheless, the majority of the current implementations rely on the rule-based logic, thus restricting the capabilities of any intelligent, ML powered zonal optimization. IoT-based laboratory systems have also covered security and reliability. Other scholars have proposed safe automation systems to minimize unauthorized accessibility and also to ensure the safety of the lab equipment in use in the engineering colleges [8]. Although these frameworks enhance system but, in most cases, they lack adaptive perception to the intelligent decision-making. Scientific and experimental laboratories also examined autonomous feedback loops in particular, with the advanced research in automation. These systems repeatedly process sensor data, and intelligent models, and adjust operating parameters with only a little human intervention [12]. In spite of these advancements in automated control, the authors indicate that human intervention remains to be done to avoid abuse and guarantee the seamless performance in the institutional contexts [9]. Laboratory on a distance and digital twin style of frameworks have been discussed to improve monitoring and simulation of the systems [2], [3]. These are used to enhance data visualization and remote access but it is more specifically remote experimentation and system modelling as opposed to real time and vision-based occupancy detection to optimize energy.

III. METHODOLOGY

The suggested framework, named as Smart Lab Automation System, is a vision enabled IoT control system that would help reduce academic lab power consumption through zone wise intelligent lighting control. System overall methodology is separated into three major stages:

(A) core system design and technology, (B) real-time sensing and zone level intelligence generation, and (C) implementation of modules for automation, monitoring and control.

All computation-intensive tasks, including human detection and decision-making, are executed locally on a Raspberry Pi 5 edge device, ensuring low latency, offline operability, and data privacy. The system leverages a lightweight YOLO based deep learning model for real-time computer vision based occupancy detection.

Live video input is captured using a camera module installed in the laboratory. Frames are pre-processed and passed to the detection pipeline, where human presence is identified. Detected individuals are spatially mapped to predefined laboratory zones. Based on zone occupancy status, lighting decisions are executed dynamically. System state, energy data, and control parameters are logged continuously for analysis and visualization.

A. Core System Architecture and Technologies

Smart Lab Automation System is designed to provide a lighting system that is intelligent, energy saving, and more intelligent in the use of computer vision, IoT-based switching, and web-based monitoring interface, located in the campus laboratory spaces. The main edge device is a Raspberry Pi that is in the center of the setup. It carries out image processing, performs the control algorithm, and the communication with the hardware components connected.

The cameras are analyzed in real time to identify people inside the lab by analyzing the video using a YOLO-based human detection model. Every identified individual is redirected to a designated area, and a policy-based algorithm determines whether lights in the particular area should be on or off. The lights are operated by relay modules which are connected to the Raspberry Pi helper pins in order to direct and effective energy consumption. The logs of energy usage are constantly recorded and saved to lightweight JSON files to be analyzed and tracked to be efficient.

A Flask and Socket.IO web dashboard can be used to offer live system status, manual control options, and configuration settings via HTTP and WebSocket communication. The general configuration saves on the energy wastage but is also practical and saleable to small laboratories as well as to campus facilities.

The proposed system shown in Figure 1 is structured into multiple functional layers: (i) real-time human detection using YOLO, (ii) zone-level occupancy management through the Zone Manager, (iii) rule-based lighting decision logic, (iv) GPIO-based relay actuation for physical switching, and (v) a web-based monitoring and control interface supported by persistent JSON data storage.

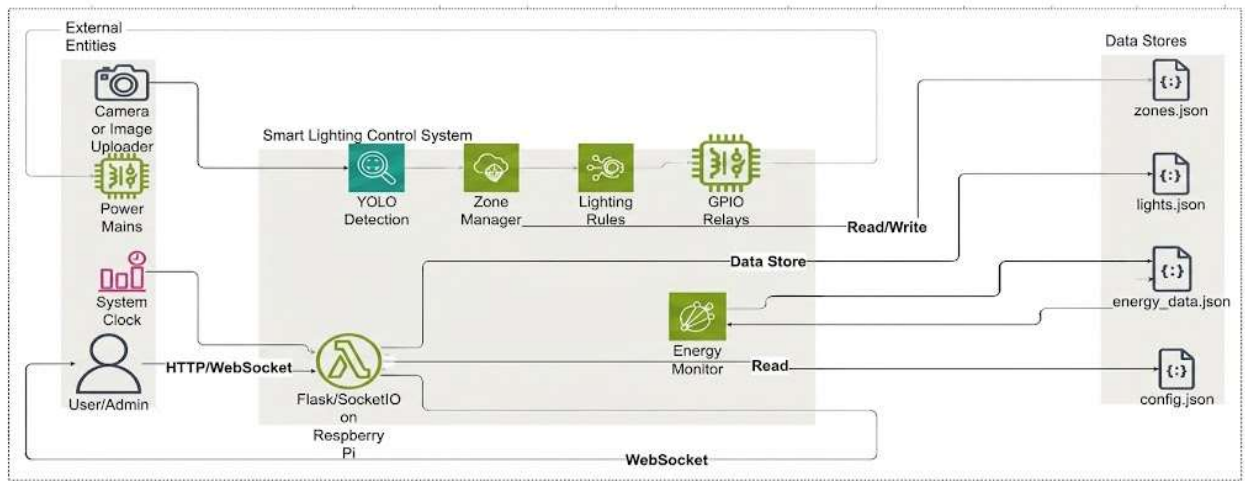


Figure 1: System data flow Architecture

There is a special Energy Monitor module where the data on lighting activity and power usage is logged. The system works with a variety of data stores in a form of JSON such as zone definitions (zones.json), lighting settings (lights.json), energy logs (energy_data.json), and system settings (config.json). The read and write functions are carried out dynamically to ensure the consistency of systems in real time. The system runs in a serial pipe, with the camera input going through the YOLO detection block, occupancy determined by the Zone Manager, lighting by the Lighting Rules block and the actual actuation by relays connected to the GPIO with the Flask server handling real-time communication and data synchronization. The Raspberry Pi 5 is used as the edge computing core of the system architecture that combines open-source hardware and software components in order to support real-time detection, decision-making, and control. It executes the computer vision model, zone-level logic of lights, and communication inside the modules as well as direct access

to hardware via the GPIO. In the case of occupancy detection, live-frame camera images are processed on the device using a deep learning model developed using the YOLO approach. The model recognizes the location of human presence and creates the bounding box co-ordinates, which are utilized in zone mapping. Relay modules are used to access the lighting control, and they are connected with the GPS of Raspberry Pi. The relays are associated with the separate zones of a laboratory and provide the possibility of a local and independent switching of lighting circuits. The backend server is a Flask based server which handles system logic, configuration parameters and inter-module communication. Socket.IO is used to provide real-time interaction between the system and the user interface where the system will update in both directions, using both HTTP and WebSocket protocols. The parameters of a system, a zone, or lights, energy logs are saved on the local drive as integrated lightweight JSON files, which are fast to access, easy to modify, and can be easily implemented

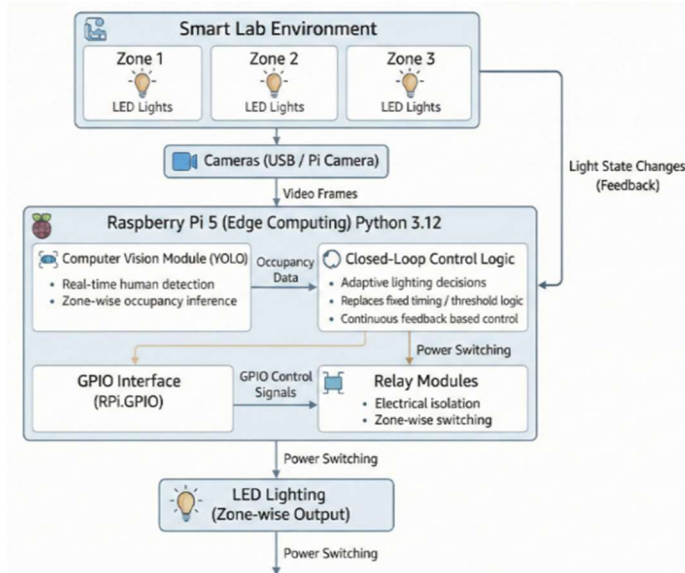


Figure 2: System Design and System Prototype

The system runs on a closed-loop control system as shown in Figure 2 wherein the data available in the camera

and other supporting sensing elements are analyzed continuously and the decisions made regarding the lighting

are immediately implemented via the networked hardware actuators. There are several control strategies that were experimented during initial testing, such as fixed time and threshold based. The behavior of these methods under variability in occupancy with a variable change deferred the choice of the closed loop strategy. The operation of the system is also mostly automatic and a web-based dashboard is also available to enable administrators to monitor the energy consumption, system status and also manually perform on demand. Configuration data and energy logs are stored in lightweight, JSON-based storage to be updated live, and to be customized easily to suit the specific system. The interaction between system components is managed by HTTP and WebSocket protocols to guarantee real time status and the need to be able to interfere manually.

B. Zone Definition and Occupancy Intelligence

To achieve fine-grained control, the laboratory space is logically divided into multiple independent zones, each mapped to a specific lighting unit. Zone boundaries are defined using spatial coordinates corresponding to the camera's field of view and are stored in a configuration file.

Regarding occupancy detection, the system utilizes centroid-based spatial analysis rather than simple presence checks. Specifically, a zone is flagged as occupied whenever the calculated center of a detected individual's bounding box falls within the defined limits. To track these changes, a Zone Manager module continuously updates occupancy states and timestamps, effectively monitoring dynamic movement within the lab also to prevent rapid switching due to transient movement, a stability control

mechanism is applied where a delay-based deactivation strategy is applied. Lights are turned off only after a zone remains unoccupied for a predefined duration.

C. Module-Specific Implementation

The system is broken into self-contained functional components so as to ensure modularity as well as to facilitate future expansion. It begins with human detection whereby video frames are obtained with the camera and processed through the YOLO model. The only detections that are considered are those over a set confidence level and this excludes false positives.

The confirmed foundations are sent to the Zone Manager, where people will be mapped to established spatial areas and occupancy states will be continuously updated. Based on this knowledge, the control module will be used to identify that the lighting changes are necessary, and the respective relay circuits are then activated via the Raspberry Pi using the GPIO interface. A component of energy monitoring works together with the control logic, which logs the lighting timings and switching instances. These records are subsequently utilized to gauge the efficiency of the system and measure the performance against the traditional light systems. A web interface is used as a visualization of the status of occupancy and lighting in zones in real-time. It also enables administrators to override automatic decisions in instances of the need of manual intervention.

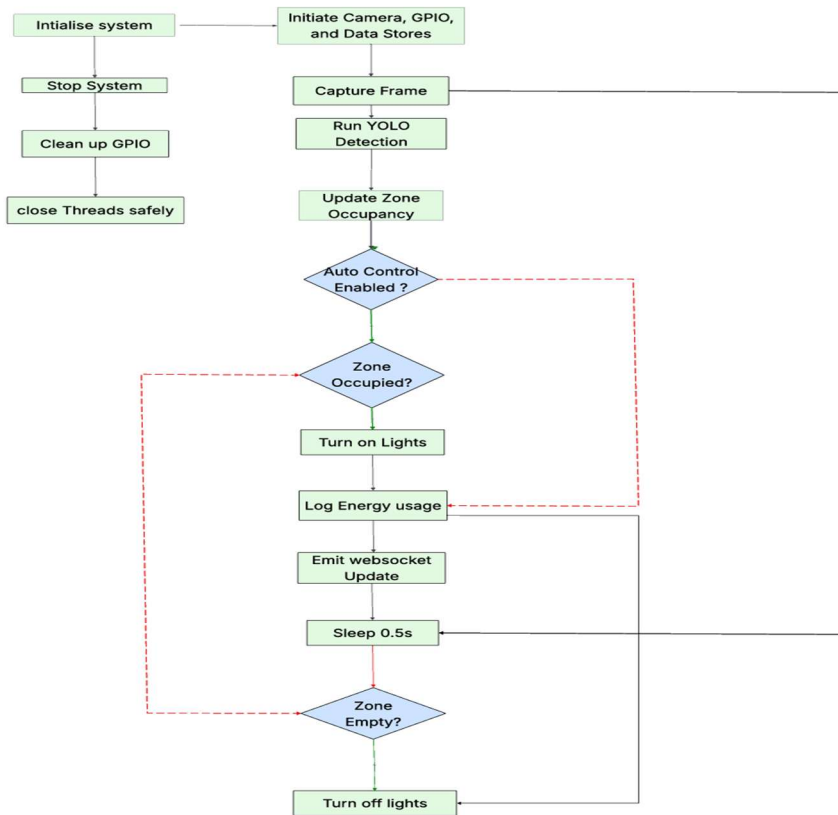


Figure 3 : System Workflow

The Figure 3 shows the operational process of the Smart Lab Automation System, including the system initialisation, continuous monitoring, decision-making processes, and safe shutdown operations. At the point of startup, the system performs initializations of major parts of the system such as the camera module, data storage files and the GPIO interface of the Raspberry Pi. This also makes sure that real-time monitoring and control are configured properly both in the hardware and software components.

After the initial phases of the system are finished the system moves into a continuity of execution. A video frame is taken and processed in every cycle by means of the human detection model built with the help of Yolo. In case individuals are observed, the corresponding zone occupancy states are updated. It is on these occupancy updates that lighting control decisions are based. In case of the activation of automatic mode, the system does an evaluation of the occupancy of the respective zone. In case of occupied areas, the relevant lighting circuits are switched on via relay modules with the help of GPIO. The monitoring and analysis of energy usage: The timestamps, and duration of energy consumption are recorded. There is a delay of 0.5 seconds between cycles to ensure responsiveness to avoid over-utilisation of the CPU. This provides a stable and efficient operation in the edge device. On the unoccupancy of a zone, the lighting is not switched off immediately. Rather, the system will ensure that a zone is vacant during a specified period and then lights are off. This is a delay mechanism ensuring the absence of frequent switching and the

increased stability of operations. This monitoring loop is dynamic and to be able to adjust to the changing occupancy conditions in the laboratory setting, the system is capable of adjusting. In both manual and automatic shutdown, the system does a controlled shutdown. Camera streams and communication threads are safely severed and the GPIO pins are reset to preclude hardware damage or corruption of data. In general, the workflow can be characterized as a closed-loop, real-time architecture of lights control which incorporates computer vision, IoT-based actuation, and web-enabled monitoring to realize energy-efficient laboratory automation.

D. Mathematical Representation of Energy Consumption

To evaluate the efficiency of the proposed smart laboratory lighting system, the electrical energy consumption of the lighting units is mathematically modelled. The total energy consumed by a lighting device depends on its power rating and the duration for which it remains active. The energy consumption can be calculated as:

$$E = P \times t \quad (1)$$

where E represents the energy consumed (Wh), P represents the power rating of the lighting unit (W), and t represents the operating time of the lighting device (hours). For a laboratory consisting of multiple lighting zones, the total energy

consumption can be expressed as the sum of the energy used by each zone.

To quantify the improvement achieved by the proposed zonal lighting control system, the percentage of energy savings compared to a conventional lighting system is calculated as:

$$\text{Energy Savings}(\%) = \frac{\{E_{\{conv\}} - E_{\{smart\}}\}}{\{E_{\{conv\}}\}} \times 100 \quad (2)$$

where E_{conv} represents the energy consumed by the conventional laboratory lighting system and E_{smart} represents the energy consumed by the proposed smart zonal lighting system. This formulation enables quantitative comparison between traditional always-on lighting approaches and the occupancy-based intelligent lighting control implemented in the proposed system.

IV. RESULTS

A. Implementation Stack & Tools

This project was developed using Python 3.12 on a Raspberry Pi 5 platform, integrating computer vision, IoT hardware control, and a web-based monitoring system the Raspberry Pi 5 functions as the edge computing platform that processes detection and control tasks locally; Python 3.12 is used as the primary programming language to integrate machine learning, hardware control, and web services; the YOLO model performs real-time human detection from live video streams; a USB or Pi Camera Module captures continuous visual input for occupancy analysis; RPi GPIO enables direct hardware control through the Raspberry Pi's GPIO pins; relay modules act as switching devices to control electrical lighting circuits; LEDs represent zone-wise lighting units for controlled energy usage; the Flask backend server manages application logic and system configuration; Flask-SocketIO provides real-time bidirectional communication between the server and dashboard; the web interface (HTML, CSS, JavaScript) delivers monitoring, visualization, and manual control capabilities; JSON files store zone definitions, configurations, and energy logs in a lightweight format; and power sensors with a custom logging module track and record energy consumption for performance analysis and optimization.

B. System Interface and Module Results

1. Zonal live camera feed interface

The Zonal Live Camera Feed is a display of the camera stream in real-time, with zone lines used which are drawn on the frame. All the zones are different regions of lighting in the laboratory which are denoted by different colored outlines. The YOLO model operates in a continuous mode to detect human beings. Upon identifying a person, there is a bounding box that is drawn around the person with a label. This allows one to find out whether the position being detected match with the spatial zones defined properly. The occupancy of all zones

varies as people move in the frame. As the person gets into a zone, the system changes its state to occupied. On making the area empty, the state represents the change following the specified delay period. These updates may be viewed right through the interface. On top of the video feed, there is a summary panel that shows such important parameters as the number of active lights, individuals detected, and power used. The updates on the values are automatically provided as a result of the communication with the backend server via WebSocket. The control panel has the option of automatic and manual functions. Testing was done using manual controls to ensure that the occupancy changes recorded in the video stream were corresponding to the activation of relays.

2. Zone Definition Interface

The Zone Definition interface gives the ability to set spatial regions based on a snapshot of captured camera. The central panel has the image feed with semi-transparent overlays, which indicate specified zones. There are also drawing modes of polygon and rectangle that one can use to mark certain parts of the image. The name given to each zone is done using an input field, and the zone will be saved. This enables it to determine individual lighting areas in the system setup. The existing areas and their status are indicated as well as the boundaries on the "Existing Zones" panel on the right. When the layout of the laboratories is altered, the changes can be modified using options like Take Snapshot and Create Zone. A configuration file contains the zone coordinates and is read by the Zone Manager to give detected people the location of lighting they belong to.

The configuration layer has nothing in common with detection and control modules.

3. Light control module interface

The Light Control interface offers direct control of single lighting units in the system. The lights are represented by control cards indicating their current status (on or off), brightness level, power rating, assigned GPIO pin and operating mode. In the process of configuration, every lighting unit is assigned each of the separate GPIO-pins on the Raspberry Pi. The control module can use this mapping to turn on or turn off the associated relay automatically when it detects occupancy, but also the interface can be used to turn on occupancy detection. When it is in manual mode, a slider control can be used to set the level of brightness. With automatic mode, decisions regarding lighting are made based on whether the occupancy is present or not as indicated by the detection and zone management modules. The system status panel on the right-hand side shows aggregate data such as number of lights, current lights, total power capacity and the current power consumption. These values change dynamically based on the activity of the relay and the state of the light. The Quick Actions area offers system level command like switching on all lights, switching off all lights or even switching on automatic mode. These were the controls that were employed during testing to ensure proper relaying response and synchronization of interface commands and hardware actuation. This interface enables one to check the

logic of switching, brightness control using GPIO and generally the lighting state in the laboratory.

C. Analysis and Output

The efficiency of the Smart Lab Zonal Lighting Control System proposed was tested with the use of 5 V DC LEDs that are managed with the help of the relay modules connected to the Raspberry Pi. The experimental test was done with the help of three spatial zones that were independent and defined in the laboratory. These zones were defined through the interface of the Zone Definition and connected to three independent lighting units of 5V DC.

There was a 1-to-1 association of each zone and a specific relay that was activated or deactivated using a 1-to-1 relationship with the GPIO. This setup enabled the comparisons of the traditional full-room lighting and occupancy-controlled zonal lighting. In the traditional installation, the three lights were kept on at all times during the laboratory time leading to a measured 24 Wh of daily energy use. The proposed zonal control system was also activated, so the lighting activation was based on occupancy in each of the predetermined zones in real time. The occupied areas were the only ones that were bright, as zones worked separately. This used up a total of 9.5Wh of energy/day.

Table 1: Energy Consumption and Savings Comparison of Conventional and Smart Zonal Lighting Systems

Parameter	Conventional Lighting System	Smart Zonal Lighting System
Supply Voltage	5 V DC	5 V DC
Lighting Load	3 LEDs (1 per zone)	3 LEDs (1 per zone)
Control Method	Manual/ Always ON	Relay-based, ML-controlled
Total Daily Energy Consumption	24 Wh	9.5 Wh
Daily Energy Saving	0 Wh	14.5 Wh
Energy Consumption (10 Days)	0.72 kWh	0.285 kWh
Total Energy Saved (10 Days)	0 Wh	0.435 kWh
Percentage Energy Savings	0 Wh	≈ 60%

Energy values were calculated based on the recorded ON duration of each lighting unit and its rated power. Over a 10-day observation period, the conventional configuration consumed 0.72 kWh, while the zonal system consumed 0.285 kWh. The total reduction of 0.435 kWh corresponds to approximately 60% energy savings. The results indicate that

defining independent zones and controlling them separately plays a key role in reducing idle lighting time in unoccupied areas.

V. CONCLUSION

The Smart Lab Automation System is a system that combines computer vision and IoT based control to overcome the issue of inefficiency in energy consumption within laboratory settings. The connection between the YOLO-based human detection and the zone-controlled lighting can be used to activate the lighting depending on the demand and not on the pre-programmed schedules or manual control. The edge computing platform was the Raspberry Pi 5 which was utilized to execute real-time detection and hardware control using GPIO without cloud processing.

The back-end system is done in Flask with Socket.IO and it handles interactions among components of detection, control and interface. A web-based dashboard is used to see the occupancy and energy consumption, and configuration information is stored in JSON files and can be adjusted to suit the system. The startup and shutdown processes were also carried out carefully in order to maintain stable operation and safeguard hardware parts. The effectiveness of the experiment revealed that the daily energy consumption had decreased to 24 Wh to 9.5 Wh of the prototype, which translated to about 60 percent saving compared to traditional lighting. These results suggest that zonal lighting control in educational labs can be made more energy efficient through occupancy-based lighting control.

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